

TORCH INFECTIONS

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Intrauterine infection transmitted from the mother to the baby.

Toxoplasmosis

Others : Syphilis, Zika virus, Parvo B19 virus.

Rubella.

Cytomegalovirus (CMV).

Herpes simplex

General aspects of Torch infection :

Type of maternal infection :

1. Primary :

- mother infected during pregnancy.
- m/c transmitted.

2. Secondary/reactivation :

- Latent infection reactivation in mother during pregnancy.
- Transmission of syphilis, CMV.

Time of gestation:

1st trimester (m/c) d/t :

- Organ anomalies.
- ↑ risk of congenital anomalies.
- Highest transmission risk.

Common manifestations :

Antenatal :

- IUGR : m/c.
- IUD.
- Preterm delivery.
- Abortions.

Postnatal:

- Suppression of bone marrow : Anemia, thrombocytopenia.
- Hepatosplenomegaly.
- Lymphadenopathy.
- Jaundice/abnormal LFT (Subclinical).

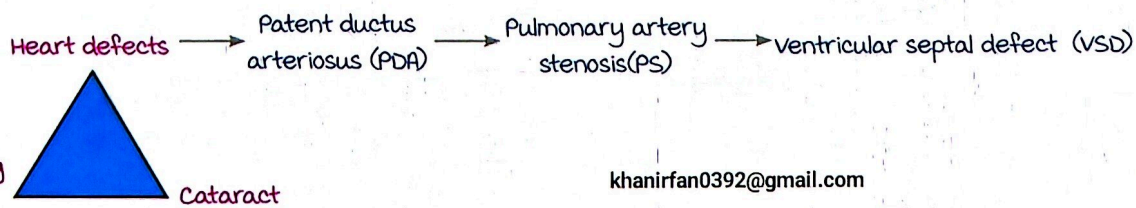
Congenital Rubella Syndrome (CRS)

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- Severe maturation infection.

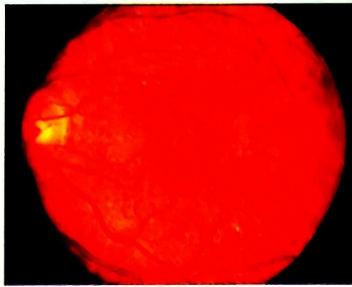
CLINICAL MANIFESTATIONS :

Clinical Triad :



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Other features :**1. Salt and pepper retinopathy :**

- Black and white spots on retina.
- m/c ocular finding.

2. Blueberry muffin lesion :

- Purpuric spots : Non-blanching dome-shaped papules.
- Found in severe CRS.
- Represents dermal erythropoiesis.
- Non-specific: Also seen in other intrauterine infections.

Chronic manifestations :**1. Progressive rubella panencephalitis (PRP) :**

- Time of onset : School going age.
- Features : Similar to SSPE (measles).
 - Poor scholastic performance.
 - Seizures.
 - Intellectual impairment.

2. Adult-onset Diabetes.**3. Hypothyroidism.****Diagnosis :**

Histology : Fetal IgM antibodies against rubella (maternal IgM cannot cross placenta).

- IgG not used as maternal IgG can cross the placenta.

Management :

- No treatment : Damage permanent.
- Prevention : Immunization of parent (During their childhood).

Congenital CMV

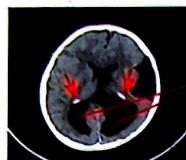
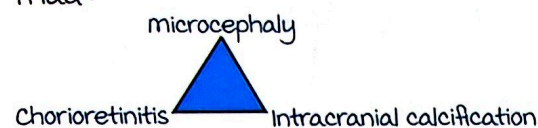
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- m/c TORCH infection.

- Transmission rate :
 - 3rd trimester (90%) → Organogenesis complete → Asymptomatic
 - 1st trimester → Organogenesis incomplete → Congenital cytomegalic inclusion disease

Clinical features :

Triad :



Periventricular calcification

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Other features :

- **SNHL** :
 - m/c long term sequelae.
 - m/c cause of non-syndromic hearing loss in children.
- **Rash** :
 - Petechiae (D/t ↓ platelets).
 - **Blueberry muffin lesion**.
- Hepatosplenomegaly.

Diagnosis :

- PCR testing (IOC) : **urine sample** > blood sample > saliva (**screening**).
- Done as early as possible.
- Serology (Low sensitivity) : IgM antibodies against CMV.

Treatment :

Ganciclovir : IV 6 weeks → Oral 6 months.

Goal :

- To prevent progression of neurological disorders.
- Decrease risk of SNHL infuture.

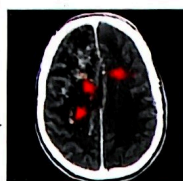
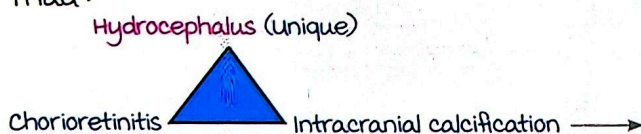
Congenital Toxoplasmosis

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- Transmission risk: highest in **3rd trimester**.

Clinical features :

Triad :



Diffuse calcification

Other manifestations :

- Low IQ.
- Deafness (SNHL).
- Some cases: **A/w microcephaly**.

Diagnosis :

- Serology :
 - IgM/IgA **ELISA** (more sensitive).
 - Immunosorbent agglutination assay (ISAGA) test.
- PCR in CSF/tissue sample (if neurological involvement +).

Treatment :

Infected child :

All cases, symptomatic and asymptomatic should be treated.

Rx : [Pyrimethamine + Sulfadiazine x 1 year] + [Folinic acid (To prevent bone marrow suppression cause by Rx)].

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Infected mothers :

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- **Spiramycin** : ↓ transmission risk.

Congenital Varicella Syndrome

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- Intrauterine transmission of herpes zoster.
- Period of risk : **2nd trimester** (13- 20 weeks of gestation).

Clinical features :

1. Cerebral atrophy.
2. microcephaly.

PERINATAL VARICELLA :

- more common than congenital varicella.

Period of highest risk : 5 days before to 2 days after delivery.

management : Immediately after birth → varicella zoster > IV Ig (Prevention).

↓ If lesions (+)

Acyclovir



3. Limb hypoplasia
4. Cicatrix (Irregular scarring, resembles burn scar)

Congenital Syphilis

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Unique feature : **Recurrent Abortion**.Period of risk : >4 months of gestation (2nd and trimester).

CLINICAL FEATURES :

Early onset : < 2 years.

1st feature :

- Rhinitis : **Snuffles**.

and feature:



Vesiculobullous rash seen in palms and soles

Pemphigus syphiliticus

3rd feature :

- Bone
 - metaphysis.
 - Osteochondritis.
 - Periostitis.
- Bone pain $\xrightarrow{\text{If severe}}$ Pseudo paralysis
+
Edema of affected limb

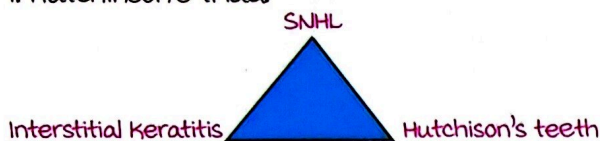


Soft tissue shadow on xray



Late onset : > 2 years.

1. Hutchinson's triad.



Notching of incisors/peg shaped incisors

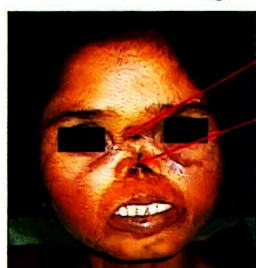
2. mulberry molars



Irregular surface of molars
(D/t presence of rudimentary cups).
- usually involves 1st molar.

3. Deformed nose :

- Preceded by : Saddle Nose deformity.



Perforations

Deformed nose (Caused by bony deformities)

4. Rhagades : Fissures around mouth/nose.

5. Saber shin : Bowing of tibia.

6. Clutton's joint : B/L knee effusion.

management :

Diagnosis : VDRL (Baby and mother).

- 4 fold ↑ intitres : Diagnostic.

Treatment :

Aqueous crystalline penicillin/ Penicillin G > 3rd generation cephalosporin : Ceftriaxone.

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Congenital Zika virus Infection

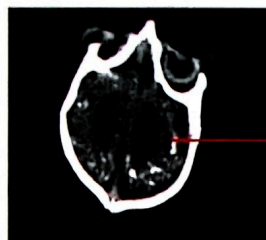
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- Vector : Aedes aegypti mosquito → Infected mother → Baby.
- Period of risk : 1st trimester.

Clinical Features :

1. Skull manifestations :

- Partial collapse of skull.
- Internal abnormalities (Eg : Overlapping of sutures).
- Brain damage :
 - microcephaly.
 - Hypertonia/spasticity.
 - Extrapyramidal manifestations: chorea, athetosis.
 - D/t basal ganglionic damage.
 - Occipital prominence.



Partial collapse of skull.

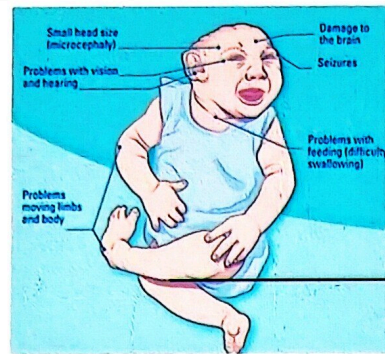


Occipital prominence

Feedback

2. Limb manifestations :

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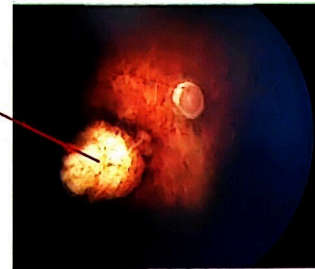


Arthrogryposis multiplex congenita

Abnormally curved limbs (d/t multiple contractures)

3. Eye manifestations :

- Coloboma.
- Pigmentary retinal mottling.
- Scar + in macula.



4. Others : SNHL.

Congenital Parvovirus B19 infection

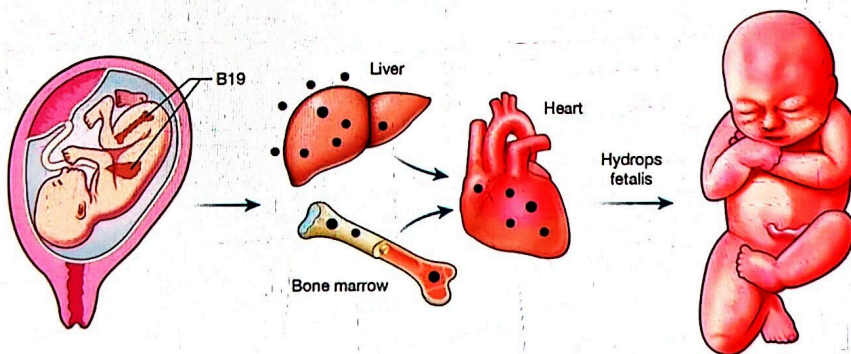
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- Transmitted across all trimesters.
 - most sensitive period of transmission : 2nd trimester.
- Tropism for erythroid lineage : Primarily at the pronormoblast stage.

Clinical Features :

1. Severe anemia → High output cardiac failure



2. Direct cytopathic effects on myocardium → myocarditis → Fetal cardiac arrest → IUD (Intrauterine death)
3. Non immune hydrops fetalis